

Introduction

Binary Search is a fundamental concept in Data Structures and Algorithms used for solving computational problems efficiently.

Definition

Binary Search refers to the method or structure used to organize, process, and manage data effectively.

Working Principle

The working of Binary Search involves step-by-step execution following a defined logic or pattern depending on the input.

Algorithm / Steps

1. Start 2. Process input 3. Apply logic 4. Return output 5. End

Example

A simple example of Binary Search helps in understanding its behavior with real inputs.

Advantages

- Easy to understand
- Efficient in many scenarios
- Widely used in applications

Disadvantages

- May be slow for large inputs
- Performance varies based on input

Time Complexity

- Best Case: $O(n)$
- Average Case: $O(n \log n)$
- Worst Case: $O(n^2)$

Space Complexity

Depends on implementation, generally $O(1)$ or $O(n)$

Applications

- Searching
- Sorting
- Data processing
- Real-world systems

Interview Questions

- Explain Binary Search
- What is its complexity?
- When to use it?

Conclusion

Binary Search is essential for mastering problem solving and technical interviews.

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